JEROME R. SCELZA, MS

BIOGRAPHICAL SKETCH - RESEARCH, INTELLECTUAL PROPERTY & SCIENTIFIC RECORD

Selected IRB / Clinical Studies | Patents | Publications | Press & External Validation

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FOCUS. Biomedical engineer, inventor and executive with a research record spanning human-subject studies, digital biomarkers, computational imaging, reproductive diagnostics, decentralized clinical research and upper-arm capillary blood collection. Selected record below is intentionally limited to research, intellectual property, publications and public validation; executive experience and employment history are provided separately.

A. IRB, CLINICAL & HUMAN-SUBJECT RESEARCH

Sleep in the Natural Environment: A Pilot Study | *Icahn School of Medicine at Mount Sinai | IRB #15-01012 | Published 2020*

Named co-author and technical contributor to a 21-participant, week-long study evaluating sleep and cognition in the natural environment using Fitbit Surge, Withings Aura, Hexoskin and Oura Ring, together with self-reported sleep, cognitive and psychological measures. Contributions included methodology, software, data curation/visualization, signal processing and data engineering. An earlier wearable-device pilot used to select study technologies was conducted under HS#15-00292.

Chaudhry FF, Danieletto M, Golden E, Scelza J, et al. Sensors. 2020;20(5):1378. DOI: 10.3390/s20051378.

Skin IaM - Exploratory Skin Inflammation Imaging Trial | *LEO Pharma / Mount Sinai | NCT03192280 | 2017*

Researcher on a registered interventional feasibility study designed to detect local inflammatory skin response before visible clinical manifestation. Prototyped the hyperspectral imaging apparatus used for longitudinal image capture and performed exploratory image processing and data analysis. The protocol combined hyperspectral imaging, thermal imaging, near-infrared spectroscopy, wearables and smartphone imaging after controlled induction of a local inflammatory response.

ClinicalTrials.gov: NCT03192280. Prior NIH-style biosketch lists role as researcher - hyperspectral apparatus prototyping, image processing and data analysis.

Lab100 - Next-Generation Preventive Health / Digital Phenotyping | *Mount Sinai Institute for Next Generation Healthcare | 2018*

Research and engineering contributor to Mount Sinai's Lab100 platform, a redesigned clinical research environment for longitudinal, multi-scale measurement of human health. The platform integrated blood testing with body composition, anthropometrics, cognition, balance, dexterity and digital assessments. Prior biosketch identifies Jerome's role in study design and data analysis. Subsequent Lab100 research operated under Mount Sinai IRB #17-00952.

University of Cambridge / Fenland Remote Blood Collection Study | *MRC Epidemiology Unit + Drawbridge Health | 2020-2022*

Large-scale decentralized blood-collection program validating OneDraw for unsupervised home use and deploying the platform for SARS-CoV-2 serosurveillance. Approximately 4,023 participants collected samples at three timepoints, generating more than 10,000 remotely collected samples; 99.9% of samples received by the laboratory were successfully processed. The program demonstrated that upper-arm capillary collection could support population-scale research outside traditional phlebotomy infrastructure.

Koulman A, Rennie KL, et al. Journal of Telemedicine and Telecare. DOI: 10.1177/1357633X221093434.

NanoDrop - Human Factors / User Preference & Device Validation | *Drawbridge Health | FDA K223826 | 2022-2024*

Executive leader during development and FDA clearance of a dual nano-lancet upper-arm blood collection device for over-the-counter use. The FDA program included a pain/preferred-method user study together with penetration depth, vacuum performance, sharps-injury prevention, biocompatibility, sterilization, shelf-life and packaging/shipping verification. Jerome Scelza is identified in the FDA 510(k) summary as Drawbridge's primary company contact and Co-CEO.

OneDraw A1C - Clinical / Method-Comparison Validation | *Drawbridge Health | FDA K183230*

Clinical and analytical evidence for the upper-arm capillary collection system included matched capillary-versus-venous HbA1c comparisons, operator-to-operator reproducibility and lot-to-lot precision. Public study materials describe a broader multi-site clinical program and strong agreement between OneDraw-collected capillary samples and conventional venous samples.

DoD PROMETHEUS / Cancer Moonshot 2.0 | *U.S. Department of Defense + Thorne / Drawbridge | 2022*

OneDraw selected for a large-scale military health research initiative using longitudinal multi-omics data to characterize health effects associated with military exposures and support prevention and earlier detection of cancer. Public program materials described deployment of thousands of OneDraw devices; Jerome was publicly quoted as Drawbridge Co-CEO on the program.

Oova - Remote Reproductive Hormone Research | *Oova / ASRM | 2022+*

Co-founder and CTO of the quantitative at-home hormone platform underlying multiple real-world reproductive-health analyses. Oova reported an ASRM 2022 analysis of 1,821 users examining hormone patterns in reproductive disorders, including PCOS. Subsequent longitudinal work used quantitative LH and PdG data from 1,233 women to characterize cycle variability and model cycle day from age and hormone measurements. The later retrospective publication states that deidentified user data were analyzed under Oova Terms of Service and did not require an IRB.

3D Anatomical Modeling & Surgical Planning | *Mount Sinai | RSNA-era research*

Engineering contributor to complex segmentation and on-site 3D printing of a severe scoliotic spine for pre-surgical planning and intra-operative visualization under John Caridi, MD. Related translational engineering included multiphysics analysis and 3D printing of personalized femoral hip implant models under Douglas Unis, MD.

Independent Biomonitoring Validation - PFAS Exposure in U.S. Veterans | *Central Arkansas Veterans Healthcare System (VA) | Published 2026*

Independent, investigator-initiated validation of the OneDraw platform for environmental-exposure biomonitoring. PFAS concentrations in dried capillary whole blood collected from the upper arm with OneDraw were compared against matched venous plasma in 133 Veterans (samples collected December 2023 - October 2024) using UHPLC-MS/MS,. Demonstrates the platform's extension from routine clinical chemistry into mass-spectrometry-based exposure science.

Journal of Exposure Science & Environmental Epidemiology. 2026. DOI: 10.1038/s41370-026-00884-5.

SODIAT-1 - Standardised and Objective Dietary Intake Assessment Tool | *Imperial College London / University of Reading / MRC Epidemiology Unit, Cambridge | ISRCTN13562899 | 2023-2025*

Randomised controlled crossover dietary-intervention trial integrating capillary blood biomarkers, urine metabolomics, wearable cameras and online dietary tools. Participants self-collected upper-arm capillary blood with OneDraw on days 1, 2 and 4 of each controlled-diet period; samples analyzed by the MRC Nutritional Biomarker Laboratory, Cambridge.

Bobokhidze E, Weech M, et al. F1000Research 2024;13:1144 (v2, 2025).

B. PUBLICATIONS & SCIENTIFIC RECORD

Sleep in the Natural Environment: A Pilot Study | *Sensors, 2020*

Chaudhry FF, Danieletto M, Golden E, Scelza J, et al. Peer-reviewed human-subject study integrating consumer wearables, mobile software, sleep metrics and cognitive assessment. DOI: 10.3390/s20051378.

Reflecting Health: Smart Mirrors for Personalized Medicine | *npj Digital Medicine, 2018*

Miotto R, Danieletto M, Scelza JR, Kidd BA, Dudley JT. Peer-reviewed work describing a functional smart-mirror platform integrating computer vision, face recognition, mood detection, respiratory-rate estimation and environmental sensing for passive health monitoring. DOI: 10.1038/s41746-018-0068-7.

Embracing the Potential of At-Home Blood Testing | *Today's Clinical Lab / Clinical Lab Products, 2022*

Scelza J. Authored industry perspective on decentralized diagnostics, patient-centric blood collection and the role of at-home sampling in expanding access, improving compliance and enabling higher-frequency longitudinal health data.

OneDraw Remote Blood Collection / Fenland Study | *Journal of Telemedicine and Telecare*

Peer-reviewed publication from the University of Cambridge / MRC program demonstrating feasibility and reliability of remote blood collection using OneDraw in a large population study. Jerome's leadership at Drawbridge supported commercialization and translation of the platform during the study period.

Longitudinal Quantitative Hormone Monitoring | *Medicina, 2023*

Hills E, Woodland MB, Divaraniya A. Study of 1,233 women using longitudinal quantitative LH and PdG measurements generated through the Oova platform to characterize menstrual-cycle variability and improve cycle-day modeling. DOI: 10.3390/medicina59071348.

Geological Processes and the Circumstellar Habitable Zone | *Earlier scientific work*

Scientific publication listed in Jerome's curriculum vitae, reflecting an early research foundation spanning engineering, physics and quantitative science.

C. PATENTS & INTELLECTUAL PROPERTY

Methods, Devices, and Systems for Detecting Analyte Levels | *Oova | Jerome R. Scelza & Aparna Divaraniya, inventors*

Patent family covering quantitative analyte detection using a consumer-facing diagnostic architecture combining assay chemistry, image acquisition and computational interpretation. The family includes U.S. publication US20210264604A1 and subsequently granted U.S. Patent US12,591,974, together with international family members including GB2591614B.

Systems and Methods for Sample Collection | *Drawbridge Health | Jerome Scelza, sole inventor | WO2023049404 / EP4405655 (EP22873668.2) / HK40114997*

Sole named inventor on Drawbridge's next-generation sample-collection patent family, filed 2022 and prosecuted internationally (PCT publication WO2023049404; European application EP4405655 published 2024 with Jerome Scelza as the only listed inventor; Hong Kong publication HK40114997). The family covers the collection architecture underlying the platform's post-OneDraw device generation. During Jerome's leadership, Drawbridge's global intellectual-property estate expanded from approximately 10 patents/pending applications to more than 80 grants and pending applications across device, collection and platform technologies.

Broader Device & Diagnostics IP Leadership | *Drawbridge Health / Oova*

Led product and technology strategy across two venture-backed diagnostics platforms built around proprietary sample acquisition and measurement technologies: quantitative smartphone-read hormone diagnostics at Oova and self-administered

upper-arm capillary blood collection at Drawbridge. Experience spans invention, freedom-to-operate strategy, patent portfolio expansion, product architecture and translation of protected technology into regulated commercial products.

D. SELECTED PRESS RELEASES & EXTERNAL VALIDATION

FDA 510(k) Clearance - NanoDrop At-Home Blood Sampling Device | *2024*
Drawbridge announced FDA 510(k) clearance of NanoDrop (K223826) for over-the-counter use - the first OTC blood-lancing device labeled for the upper arm cleared in the United States. Public coverage quotes Jerome as Co-CEO on decentralized clinical trials, remote populations and next-generation at-home blood collection.

NanoDrop Receives CE Mark | *2023*
Thorne / Drawbridge announced European CE Mark certification for the dual nano-lancet platform, supporting at-home blood sampling applications and international commercialization.

OneDraw Named Best Overall Medical Device Product | *2023*
MedTech Breakthrough recognized OneDraw as Best Overall Medical Device Product during Jerome's leadership of Drawbridge.

Department of Defense Cancer Moonshot 2.0 - PROMETHEUS | *2022*
Thorne announced OneDraw's use in the DoD PROMETHEUS initiative, with thousands of devices supporting longitudinal multi-omics research in U.S. service members. Jerome was quoted as Drawbridge Co-CEO.

OneDraw Receives PMDA Certification in Japan | *2022*
Thorne announced Japanese medical-device certification for OneDraw through the Pharmaceutical and Medical Devices Agency, expanding the platform's global regulatory footprint. Jerome was quoted on access and global deployment.

Large-Scale Cambridge Study Confirms Remote OneDraw Reliability | *2022*
Public release highlighted approximately 4,000 participants, more than 10,000 remotely collected samples and 99.9% successful processing of samples received by the laboratory.

Thorne / GenTegra Next-Generation Whole-Blood DNA Collaboration | *2022*
Collaboration paired OneDraw's dried-blood collection matrix with GenTegra Active Chemical Protection and GenSolve extraction technology to support preservation and recovery of DNA from whole-blood matrices. Jerome was quoted as Drawbridge Co-CEO.

Thorne Acquires Majority Interest in Drawbridge Health | *2021*
Strategic acquisition announcement positioned Drawbridge's blood collection technology within Thorne's personalized-health platform and named Jerome as Co-CEO.

Oova Raises $10.3M and Expands At-Home Fertility Platform | *2023*
TechCrunch coverage of Oova's financing and membership expansion highlighted the quantitative at-home fertility testing platform Jerome co-founded and helped develop.

Oova Seed Financing / Product Launch | *2021*
TechCrunch covered Oova's $1.2M seed financing and launch of its quantitative at-home hormone testing platform measuring LH and progesterone metabolite trends.

Novel Device Offers At-Home Fertility Monitoring | *Oova / Mount Sinai*
Clinical coverage highlighted Oova as a Mount Sinai spinout and its use of quantitative at-home hormone testing to personalize reproductive-health monitoring.

Drawbridge Announces Partnerships to Drive Expansion in Japan | *Drawbridge Health / Sumitomo Dainippon Pharma* | *2020*
Strategic partnerships, including with Sumitomo Dainippon Pharma, to bring OneDraw to Japan - the groundwork for OneDraw's 2022 PMDA certification.

Parexel / Synexa Life Sciences / Drawbridge COVID-19 Serology Collaboration | *Parexel International* | *2020*
Tripartite collaboration pairing OneDraw remote blood collection with Synexa's serology assays and Parexel's clinical-trial platform to enable distributed sampling in pivotal COVID-19 trials and sero-prevalence studies.

SOURCE RECORD

Selected sources include FDA 510(k) records K183230 and K223826; ClinicalTrials.gov NCT03192280; ISRCTN13562899; EPO/WIPO patent records; peer-reviewed publications in Sensors, npj Digital Medicine, Journal of Telemedicine and Telecare, Medicina, Journal of Exposure Science & Environmental Epidemiology and F1000Research; Mount Sinai research records; University of Cambridge / MRC study materials; Oova research and patent records; and public releases from Drawbridge Health, Thorne HealthTech and related institutional/industry sources.

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